BIOMEDICAL LIMITED ILORIN	STANDARD TESTING PROCEDURE
	TITLE: STANDARD CHEMICAL TESTING PROCEDURE FOR FLEXRON BLOOD TONIC (200ml)

DEPARTMENT:	**QUALITY**	**SECTION:**	**QUALITY CONTROL**
STP NO.:	**BML/QA/STP/FPS/016/F01**	**PAGE NO.:**	**Page 1 of 7**
STP Ref.:	**BML-QDG-QA-105**	**REF. BOOK**	**BP 2024/IN-HOUSE**
EFFECTIVE DATE:	**04/01/2024**	**REVISION NO:**	**01**
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S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Ferric Ammonium Citrate** **Label Claim: 200mg/5ml**	**METHOD 1:** Reagents: a. 20% solution of Citric Acid b. Strong Ammonia Solution c. 80% Thioglycollic Acid d. Concentrated Hydrochloric Acid ***Standard solution:*** Weigh accurately 200mg of Ferric Ammonium Citrate CRS, add 10ml of concentrated hydrochloric acid and dilute to 100ml with water. Futher dilute 5ml to 100ml with water. ***Sample solution:*** To appropriate quantity of the sample, add 10ml of concentrated hydrochloric acid and then dilute with water. **Procedure:**

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	**QUALITY CONTROL SENIOR ANALYST**		
REVIEWED BY:	**QUALITY CONTROL HEAD**		
APPROVED BY:	**QUALITY ASSURANCE MANAGER**		

To 10ml each of the sample and standard solutions, add 2ml of Citric Acid solution, 0.1ml of Thioglycollic acid and 5ml of strong Ammonia solution, dilute to 25ml with water.

Measure the Absorbance of both solutions at 525nm against reagent blank. This will give Total Quantity (mg) of Iron in the sample.

% Assay of Ferric Ammonium Citrate = $\frac{absorbance of sample}{absorbance of standard}$ x 100

Note: 100 mg of Ferric Ammonium Citrate contains 21.5mg of elemental Iron.

METHOD 2

Digest accurately weighed quantity of the sample with 4-5ml of concentrated hydrochloric acid. Transfer to 250ml flask with the help of water. Add 15ml of Potassium Iodide solution (10%w/v). Keep in the dark for 5 min and titrate the liberated iodine with 0.1N Sodium thiosulphate using starch paste as indicator, added near the end titration. Each ml of 0.1N Sodium thiosulphate is equivalent to 5.585mg of Iron. Calculate the contents of FAC in the sample considering each 100mg of FAC equivalent to 21.5mg of Iron.

$$\%\,\boldsymbol{Assay} = \frac{(Average\ Titre)\ x\ Standard\ Factor\ x\ \ 21.5}{200}\ x\ 100$$

REFERENCE:

		Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
3.	**Folic Acid** **Label Claim:** **0.5mg/5ml**	**METHOD 1:** This method is based on the reaction of Folic acid with formaldehyde to yield a coloured **Reagents:** a. 0.01N Sodium Hydroxide b. 1M Sulphuric Acid ***Standard solution:*** 100mcg/ml of Folic Acid in 0.01N Sodium Hydroxide. ***Sample solution:*** To appropriate quantity of the sample equivalent to 5mg of Folic acid and extract with 25ml of 0.01N Sodium Hydroxide by shaking for 15 minutes, filter and dilute to 50ml with water. Procedure**:** To 10ml each of the sample and standard solutions, add 13ml of 1M Sulphuric acid and 3ml of formaldehyde solution. Close the tube with rubber stopper in which a hypodermic needle has been inserted. Heat on a boiling water – bath for 1 hour, cool and dilute to 25ml

		with water. Measure the absorbance of sample and standard at maximum at about 360nm against a reagent blank prepared by adding 13ml of 1M Sulphuric acid to 10ml of sample solution and diluting to 25ml with water. **% Assay of Folic Acid** = $\frac{absorbance of sample}{absorbance of standard}$ x 100 ***REFERENCE:*** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 4
4.	**Microbial Enumeration Tests**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method. - Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette. - Open the lid of the prepared media, flame its mouth. - Pour about 15 mL of sterilized molten media at the appropriate temperature above the sample. - Close the lid of the plate then mix the sample in the media properly by gently swirling the plate. The plate is generally swirled in an "S" or "8" shape. - Allow the media to completely solidify. - Incubate the plate in an inverted position under suitable incubation conditions

(Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days.

TEST FOR ABSENCE OF PATHOGEN

- Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth
- Transfer 10ml into 90ml of Tryptone soya broth
- Incubate at 30-35^{0}c for 18-24hours
- Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours
- Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c
- For negative control, proceeding same as above except without sample, streak a portion of incubated sterile MacConkey broth medium on the surface of Eosin Methylene Blue Agar.
- Incubate plates at 30 to 35° for 48 hours.

A. Result Interpretation

- Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit.

- If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.
- However, if there are different genera or species producing similar types of colonies, it is required to carry out identification test.
- If the colonies are of different morphology, then we can conclude that the sample contained a mixed population.

Count the colonies and calculate CFU/mL by using the formula:

$$\textbf{CFU/mL} = \frac{\textbf{Total number of colonies obtained x dilution factor}}{\textbf{Volume of sample used (aliquot)}}$$

- This will give the total number of viable microbial cells present in the given sample.
- For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated.
- If the number of the colony is less than 20, it is suggested to use the sample of lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use the sample of higher dilution on successive repeats.
- The result should be recorded following the Standard testing procedure SOP **BML-QDG-QA-105**
- If the colonies are fused or the whole plate is covered with a single colony, then

		report as “too numerous to count” (TNTC) and repeat the process of taking the sample at a higher dilution.

CURRENT REVI	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

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PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
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